

mistake was made?

41. Suppose a base 12 number is converted into a base 10 number, and one of the digits is A. Was an error committed? How do you know?
42. Suppose a base 10 number is converted into a base 5 number and one of the digits is 6. Was an error committed? How do you know?
43. Suppose a base 2 number is converted into a base 10 number, and the base 10 number has more digits than the base 2 number. Was an error committed? How do you know?
44. Suppose a base 16 number is converted to base 2, and the base 2 number has fewer digits than the base 16 number. Was an error committed? How do you know?

For the following exercises, convert the Hindu-Arabic number into a Mayan number.

45. 25
46. 71
47. 400
48. 723

The Babylonian system used base 60. To convert from Hindu-Arabic numbers into Babylonian numbers, the process for converting from base 10 to a different base would be done first. Then, the results found in the conversion process would be changed to Babylonian numerals. This process is similar to the one for Mayan numbers.

The Babylonian system used base 60. To convert from Hindu-Arabic numbers into Babylonian numbers, the process for converting from base 10 to a different base would be done first. Then, the results found in the conversion process would be changed to Babylonian numerals. This process is similar to the one for Mayan numbers. For the following exercises, convert the Hindu-Arabic number to a Babylonian number.

49. 67
50. 135
51. 781
52. 10,952

4.4 Addition and Subtraction in Base Systems

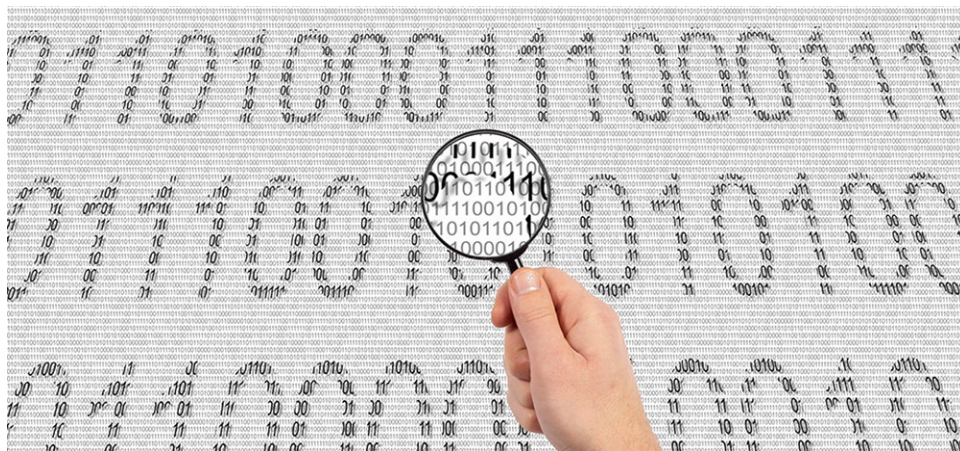


Figure 4.5 All information in computers is represented by 0's and 1's, including quantity, which means computers use Base 2 for arithmetic. (credit: modification of work "Magnifying glass and binary code" by Marco Verch Professional Photographer/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Add and subtract in bases 2–9 and 12.
2. Identify errors in adding and subtracting in bases 2–9 and 12.

Once we decide on a system for counting, we need to establish rules for combining the numbers we're using. This begins with the rules for addition and subtraction. We are familiar with base 10 arithmetic, such as $2 + 5 = 7$ or $3 \times 5 = 15$. How does that change if we instead use a different base? A larger base? A smaller one? In particular, computers use base 2 for all number representation. When your calculator adds or subtracts, multiplies or divides, it uses base 2. This is because

the circuitry recognizes only two things, high current and low current, which means the system is uses only has two symbols. Which is what base 2 is.

In this section, we use addition and subtraction in bases other than 10 by referencing the processes of base 10, but applied to a new base system.

Addition in Bases Other Than Base 10

Now that we understand what it means for numbers to be expressed in a base other than 10, we can look at arithmetic using other bases, starting with addition. When you think back to when you first learned addition, it is very likely you learned the addition table. Once you knew the addition table, you moved on to addition of numbers with more than one digit. The same process holds for addition in other bases. We begin with an addition table, and then move on to adding numbers with two or more digits.

We worked with base 6 earlier, and have the numbers in base 6 up to 100_6 . Using that table of values, we can create the base 6 addition table.

Here's the beginning of the base 6 addition table:

+	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	?
2	2	3	4	5	?	?
3	3	4	5	?	?	?
4	4	5	?	?	?	?
5	5	?	?	?	?	?

Many of the cells are not filled out. The ones filled in are values that never get past 5, which is the largest legal symbol in base 6, so they are acceptable symbols. But what do we do with $5 + 3$ in base 6? We can't represent the answer as "8" since "8" is not a symbol available to us. Let's go back to the list of numbers we have for base 6.

0	1	2	3	4	5
10	11	12	13	14	15
20	21	22	23	24	25
30	31	32	33	34	35
40	41	42	43	44	45
50	51	52	53	54	55

So, what is $5 + 1$ equal to in base 6? Well, start at the 5, and jump ahead one step. You land on 10.

0	1	2	3	4	5
10	11	12	13	14	15
20	21	22	23	24	25
30	31	32	33	34	35
40	41	42	43	44	45
50	51	52	53	54	55

0	1	2	3	4	5
10	11	12	13	14	15
20	21	22	23	24	25
30	31	32	33	34	35
40	41	42	43	44	45
50	51	52	53	54	55

The 10 is one step past the 5

This means that, in base 6, $5 + 1 = 10$.

So, what is $5 + 2$ in base 6? Well, $5 + 2 = 5 + 1 + 1$, so $10 + 1 \dots$ jump one more space and you land on 11. So, $5 + 2 = 11$ in base 6.

0	1	2	3	4	5
10	11	12	13	14	15
20	21	22	23	24	25
30	31	32	33	34	35
40	41	42	43	44	45
50	51	52	53	54	55

And so it goes. Using that process, stepping one more along the list, we can fill in the remainder of the base 6 addition table ([Table 4.4](#)).

+	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	10
2	2	3	4	5	10	11
3	3	4	5	10	11	12

Table 4.4 Base 6 Addition Table

4	4	5	10	11	12	13
5	5	10	11	12	13	14

Table 4.4 Base 6 Addition Table

With this table, and with our understanding of “carrying the one,” we can then use the addition table to do addition in base 6 for numbers with two or more digits, using the same processes you learned for addition when you did it by hand.

EXAMPLE 4.28**Adding in Base 6**

Calculate $251_6 + 133_6$.

✓ **Solution**

Step 1: Let's set up the addition using columns.

	2	5	1
+	1	3	3

Step 2: Let's do the one's place first. According to the base 6 addition table (Table 4.4), $1 + 3 = 4$.

	2	5	1
+	1	3	3
			4

Step 3: Now, we do the “tens” place (it's really the sixes place). According to the base 6 addition table (Table 4.4), we have $5 + 3 = 12$. So, like in base 10, we use the 2 and carry the 1.

	1		
	2	5	1
+	1	3	3
		2	4

Step 4: Now the “hundreds” place (really, thirty-sixes place). There, we have $1 + 2 + 1 = 3 + 1 = 4$.

	1		
	2	5	1
+	1	3	3
	4	2	4

So, $251_6 + 133_6 = 424_6$.

As you can see, the process is the same as when you learned base 10 addition, just a different symbol set.

> **YOUR TURN 4.28**

1. Calculate $453_6 + 345_6$.

EXAMPLE 4.29

Creating an Addition Table for a Base Lower Than 10

1. Create the addition table for base 7.
2. Create the addition table for base 2.

✓ **Solution**

1. We begin with the table below.

+	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	2	3	4	5	6	
2	2	3	4	5	6		
3	3	4	5	6			
4	4	5	6				
5	5	6					
6	6						

In base 7, the number that follows 6 is 10 (since we've run out of symbols!). So, $6_7 + 1_7 = 10_7$. Once that is established, $6_7 + 2_7$ will be two numbers past 6, which is 11 in base 7.

+	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	2	3	4	5	6	10
2	2	3	4	5	6		11
3	3	4	5	6			
4	4	5	6				
5	5	6					
6	6	10	11				

Continuing, we can fill in the rows as we would in base 10, but being aware that we are working in base 7 ([Table 4.5](#)).

+	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	2	3	4	5	6	10
2	2	3	4	5	6	10	11
3	3	4	5	6	10	11	12
4	4	5	6	10	11	12	13
5	5	6	10	11	12	13	14
6	6	10	11	12	13	14	15

Table 4.5 Base 7 Addition Table

2. We revisit base 2 here. Begin with the table:

+	0	1
0	0	1
1	1	

Table 4.6
Base 2
Addition
Table

In base 2, the number that follows 1 is 10 (since we’ve run out of symbols!). So, $1_2 + 1_2 = 10_2$. The complete table for base two then is below.

+	0	1
0	0	1
1	1	10

This demonstrates that the rules necessary for base 2 addition are as small as possible: four rules.

> **YOUR TURN 4.29**

1. Create the addition table for base 4.

To summarize the creation of the addition tables for a given base, do the following.

Step 1: Set up the table.

Step 2: Fill in all the additions that use the “legal” symbols for the base. The diagonal that goes from upper left to lower right that is immediately next to the filled boxes all get the value 10, regardless of base.

Step 3: Enter the values that are in the “teens.” This can all be done on one table without creating multiple copies of previously done work.

EXAMPLE 4.30

Adding in Base 7

Calculate $536_7 + 433_7$.

✓ Solution

Step 1: Let's set up the addition using columns.

	5	3	6
+	4	3	3

Step 2: Let's do the one's place first. According to the base 7 addition table in the solution for [Example 4.29](#), $6 + 3 = 12$. We will carry the 1.

		1	
	5	3	6
+	4	3	3
			2

Step 3: Now, we do the “tens” place (it's really the sevens place). According to the base 7 addition table in the solution for [Example 4.29](#), we have $1 + 3 + 3 = 10$. So, like in base 10, we use the 0 and carry the 1.

	1		
	5	3	6
+	4	3	3
		0	2

Step 4: Now the “hundreds” place (really, forty-ninths place). There, we have $1 + 5 + 4 = 6 + 4 = 13$.

	1		
	5	3	6
+	4	3	3
1	3	0	2

So, $536_7 + 433_7 = 1302_7$.

> YOUR TURN 4.30

1. Calculate $461_7 + 142_7$.

As seen previously, when performing addition in another base, set up the problem exactly as you would for addition in base 10. At each step, check the addition table for the base. As in base 10 addition, move right to left, adding down the columns using the rules in the addition table. When necessary and just as in base 10, be sure to carry the 1.

EXAMPLE 4.31

Creating an Addition Table for a Base Higher Than 10

Create the addition table for base 12.

✓ Solution

Step 1: Recall, in base 12, the symbol set is 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, and B. So, the addition table begins as shown below.

+	0	1	2	3	4	5	6	7	8	9	A	B
0	0	1	2	3	4	5	6	7	8	9	A	B
1	1	2	3	4	5	6	7	8	9	A	B	
2	2	3	4	5	6	7	8	9	A	B		
3	3	4	5	6	7	8	9	A	B			
4	4	5	6	7	8	9	A	B				
5	5	6	7	8	9	A	B					
6	6	7	8	9	A	B						
7	7	8	9	A	B							
8	8	9	A	B								
9	9	A	B									
A	A	B										
B	B											

Step 2: The diagonal immediately to the right of the filled in boxes is where the 10 goes for this base.

+	0	1	2	3	4	5	6	7	8	9	A	B
0	0	1	2	3	4	5	6	7	8	9	A	B
1	1	2	3	4	5	6	7	8	9	A	B	10
2	2	3	4	5	6	7	8	9	A	B	10	

3	3	4	5	6	7	8	9	A	B	10		
4	4	5	6	7	8	9	A	B	10			
5	5	6	7	8	9	A	B	10				
6	6	7	8	9	A	B	10					
7	7	8	9	A	B	10						
8	8	9	A	B	10							
9	9	A	B	10								
A	A	B	10									
B	B	10										

Step 3: Using the pattern we're familiar with, and counting in base 12, we can fill in the other cells.

+	0	1	2	3	4	5	6	7	8	9	A	B
0	0	1	2	3	4	5	6	7	8	9	A	B
1	1	2	3	4	5	6	7	8	9	A	B	10
2	2	3	4	5	6	7	8	9	A	B	10	11
3	3	4	5	6	7	8	9	A	B	10	11	12
4	4	5	6	7	8	9	A	B	10	11	12	13
5	5	6	7	8	9	A	B	10	11	12	13	14
6	6	7	8	9	A	B	10	11	12	13	14	15
7	7	8	9	A	B	10	11	12	13	14	15	16
8	8	9	A	B	10	11	12	13	14	15	16	17
9	9	A	B	10	11	12	13	14	15	16	17	18
A	A	B	10	11	12	13	14	15	16	17	18	19
B	B	10	11	12	13	14	15	16	17	18	19	1A

Table 4.7 Base 12 addition table

Notice that the lower-right entry is $1A_{12}$, as this is the number one past 19_{12} .

> YOUR TURN 4.31

1. Create the addition table for base 14.

EXAMPLE 4.32**Adding in Base 12**

Calculate $3A7_{12} + 9BA_{12}$.

✓ Solution

Step 1: Using the process established in the earlier addition problem, set up the columns.

	3	A	7
+	9	B	A

Step 2: Using the rules from the base 12 addition table in the solution for [Example 4.31](#), and being careful to carry the 1 when necessary, we get the following:

	1	1	
	3	A	7
+	9	B	A
1	1	A	5

The ones that were carried are located over the columns.

So, $3A7_{12} + 9BA_{12} = 11A5_{12}$.

> YOUR TURN 4.32

1. Calculate $4B3_{12} + B06_{12}$.

EXAMPLE 4.33**Adding in Base 2**

We again return to base 2, the base used by computers. Calculate $1001_2 + 11011_2$.

✓ Solution

Step 1: Using the process established in the earlier addition problem, set up the columns.

		1	0	0	1
+	1	1	0	1	1

Step 2: Using the rules from the base 2 addition table in the solution for [Example 4.29](#), and being careful to carry the 1 when necessary (and shown at the top of the grid), we get the following:

	1		1	1	
		1	0	0	1
+	1	1	0	1	1
1	0	0	1	0	0

Step 3: Calculate $1001_2 + 11011_2 = 100100_2$.

So, $1001_2 + 11011_2 = 100100_2$.

> YOUR TURN 4.33

1. Calculate $101111_2 + 1100011_2$.

Subtraction in Bases Other Than Base 10

Subtraction in bases other than base 10 follow the same processes as base 10 subtraction, but, as with addition, using the addition table for the base.

EXAMPLE 4.34

Subtracting in Base 6

Calculate $52_6 - 34_6$.

✓ Solution

Step 1: Let's set up the subtraction using columns.

	5	2
-	3	4

Step 2: Just as we might do in base 10, we borrow a 1 from the 5 for the ones digit.

	5 4	12
-	3	4

Step 3: Referring to the base 6 addition table (Table 4.4), we see that $4 + 4 = 12$, so $12_6 - 4_6$ is 4_6 .

	5 4	12
-	3	4
		4

Step 4: Now we deal with the “tens” (really, sixes) digit, $4_6 - 3_6$, which equals 1_6 according to the base 6 addition table (Table 4.4).

	5 4	12
-	3	4
	1	4

So, $52_6 - 34_6 = 14_6$.

> YOUR TURN 4.34

1. Calculate $115_6 - 43_6$.

EXAMPLE 4.35

Subtracting in Base 12

Calculate $A17_{12} - 4B3_{12}$.

✓ Solution

Step 1: Let's set up the subtraction using columns.

	A	1	7
-	4	B	3

Step 2: Even in base 12, $7_{12} - 3_{12} = 4_{12}$.

	A	1	7
-	4	B	3
			4

Step 3: Moving to the “tens” digit, we have $1_{12} - B_{12}$. Since 1 is less than B in base 12, we need to borrow a 1 from the A, just as we would for subtraction in base 10.

	A 9	11	7
-	4	B	3
			4

Step 4: According to the base 12 addition table in the solution for [Example 4.31](#), $B_{12} + 2_{12} = 11_{12}$, so $11_{12} - B_{12} = 2_{12}$.

	A 9	11	7
-	4	B	3
		2	4

Step 5: Finally, we deal with the “hundreds” digit. According to the base 12 addition table in the solution for [Example 4.31](#), $4_{12} + 5_{12} = 9_{12}$, so $9_{12} - 4_{12} = 5_{12}$.

	A 9	11	7
-	4	B	3
	5	2	4

So, $A17_{12} - 4B3_{12} = 524_{12}$.

> YOUR TURN 4.35

1. Calculate $716_{12} - 4AB_{12}$.

Errors When Adding and Subtracting in Bases Other Than Base 10

Errors when computing in bases other than 10 often involve applying base 10 rules or symbols to an arithmetic problem in a base other than base 10. The first type of error is using a symbol that is not in the symbol set for the base. For instance, if a 9 shows up when working in base 7, you know an error has happened because 9 is not a legal symbol in base 7.

EXAMPLE 4.36

Identifying an Illegal Symbol in Arithmetic in a Base Other Than Base 10

Explain the error in the following calculation:

$$15_6 + 34_6 = 49_6$$

✓ Solution

Since the problem is in base 6, the symbol set available is 0, 1, 2, 3, 4 and 5. The 9 in the answer is clearly not a legal symbol for base 6. Looking back to the base 6 addition table ([Table 4.4](#)), we see that $5_6 + 4_6 = 13_6$. Correcting the error, we see the sum is $15_6 + 34_6 = 53_6$.

> YOUR TURN 4.36

1. Explain the error in the following calculation and correct the problem:
 $133_4 + 112_4 = 245_4$

The second type of error is using a base 10 rule when the numbers are not in base 10. For instance, if you are working in base 13, then $9_{13} + 9_{13}$ is not 18_{13} , even though 18 is the correct answer in base 10.

EXAMPLE 4.37**Identifying an Arithmetic Error in a Base Other Than Base 10**

Explain the error in the following calculation, and correct the error:

$$89_{12} + 76_{12} = 165_{12}$$

 Solution

If this problem was a base 10 problem, this would be the correct answer. However, in base 12, $9 + 6$ is not 15, but is instead 13. To correct this error, carefully use the [addition table for base 12](#). If properly used, the correct answer would be 143_{12} , as seen below:

		8	9
+		7	6
	1	4	3

 YOUR TURN 4.37

1. Explain the error in the following calculation, and correct the error:

$$149_{14} + 19_{14} = 168_{14}$$

Check Your Understanding

21. Determine the addition table for base 8.
22. Compute $24_6 + 53_6$.
23. Compute $35_8 - 26_8$.
24. Compute $3B_{14} + 45_{14}$.
25. Compute $A4_{12} - 9B_{12}$.
26. How do you know an error has occurred in a base 8 addition question if the answer obtained was 28_8 ?
27. What is one common error made in calculating in base 14?

**SECTION 4.4 EXERCISES**

For the following exercises, create the addition table for the given base.

1. base 5
2. base 3
3. base 16 (Hint: Use the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, and F.)
4. base 2

For the following exercises, perform the indicated base 6 operation.

5. $4_6 + 3_6$
6. $14_6 + 25_6$
7. $31_6 + 3_6$
8. $43_6 + 34_6$
9. $532_6 + 23_6$
10. $254_6 + 143_6$
11. $20_6 - 3_6$
12. $23_6 - 5_6$

For the following exercises, perform the indicated base 12 operation.

- 13. $5_{12} + 6_{12}$
- 14. $3_{12} + A_{12}$
- 15. $34_{12} + 7_{12}$
- 16. $76_{12} + B_{12}$
- 17. $59_{12} + 1A_{12}$
- 18. $A1_{12} + 36_{12}$
- 19. $53_{12} - 9_{12}$
- 20. $2B_{12} - 7_{12}$

- 21. Explain two ways to detect an error in arithmetic in bases other than base 10.
- 22. Explain the error in the following calculation: $28_{13} + 47_{13} = 75_{13}$.
- 23. Explain the error in the following calculation: $36_7 + 23_7 = 59_7$.
- 24. In base 10 addition, there are 100 addition rules plus a rule for carrying a 1. How many addition rules are there for base 6?
- 25. In base 10 addition, there are 100 addition rules plus a rule for carrying a 1. How many addition rules are there for base 14?
- 26. In base 10 addition, there are 100 addition rules plus a rule for carrying a 1. How many addition rules are there for base 2?

For the following exercises, use the addition table that you created from Exercise 4 to perform the indicated base 2 operations.

- 27. $101_2 + 111_2$
- 28. $1011_2 + 10011_2$
- 29. $11111_2 + 11111_2$
- 30. $1010101_2 + 1010101_2$

For the following exercises, use the addition table that you created from Exercise 3 to perform the indicated base 16 operations.

- 31. $29_{16} + 38_{16}$
- 32. $4D_{16} + 89_{16}$
- 33. $927_{16} + 438_{16}$
- 34. $BFA_{16} - 78E_{16}$

For the following exercises, tell how you know an error was committed without performing the operation in the given base.

- 35. $43_5 + 32_5 = 75_5$
- 36. $15_{14} + 19_{14} = 34_{14}$